CSE 20 Winter 2026
Exam Practice

1. Modeling, Sets and Functions, Algorithms In machine learning, clustering can be used to group similar data for prediction and recommendation. For example, each Netflix user's viewing history can be represented as a n-tuple indicating their preferences about movies in the database, where n is the number of movies in the database. Each element in the n-tuple indicate the user's rating of the corresponding movie: 1 indicates the person liked the movie, -1 that they didn't, and 0 that they didn't rate it one way or another.

Consider the following algorithm for determining if a user's viewing history represents strong opinions on many movies.

Determine if user's ratings tuple encode strong opinions

```
procedure opinion((r_1, ..., r_n): a n-tuple of ratings; c: a nonnegative integer)
sum := 0
for i := 1 to n
   if r_i ≠ 0
      sum := sum + 1
return sum ≥ c
```

1. When $n = 3$, describe the set of all n-tuples representing user ratings
   (a) By the roster method.
   (b) With set builder notation.
   (c) Using a recursive definition.
2. What are the possible return (output) values of this algorithm?
3. **Trace** the computation of $opinion((-1, 0, 0, 0, 1, 1, -1), 2)$.
4. Give an example of a nonnegative integer c so that, no matter which n-tuple $(r_1, \ldots, r_n)$ we consider, $opinion((r_1, \ldots, r_n), c)$ will have the same value. What value is it?

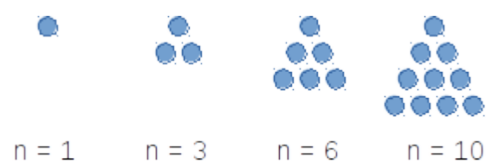
2. Sets and Functions

1. Give a recursive definition for the set $\mathbb{N}$.
2. Use the recursive definition from part (a) to give a recursive definition for the function with domain $\mathbb{N}$, codomain $\mathbb{N}$ which computes, for input i, the sum of the first i powers of 2. For example, on input 0, the function evaluates to 2^0, namely to 1. On input 1, the function evaluates to $2^0 + 2^1$, namely 3. On input 2, the function evaluates to $2^0 + 2^1 + 2^2$, namely 7.

3. Base expansions, Multiple representations

1. Compute the ternary (base 3) expansion of 28.
2. Compute the product of $(6A)_{16}$ and $(11)_{16}$, without converting either number to another base.
3. Confirm your answer for part (b) by converting $(6A)_{16}$ and $(11)_{16}$ to decimal, multiplying them, and converting the product to base 16.
4. How many bits will there be in the binary (base 2) expansion of 2020? Can you compute this without fully converting 2020 to base 2?
5. Give an example of a number that can be represented in base 2 fixed-width 3, but not in base 2 expansion.
6. Give an example of a number that can be represented in base 2 expansion, but not in base 2, fixed-width 3 expansion.
7. Give the representation of -7 in sign-magnitude width 3 and 2s complement width 3. Then do the same for width 4.
8. Give the representation of 10.5625 in fixed-width base-2 expansion with integer width 4 and fractional width 9.

4. Circuits A triangular number (or triangle number) counts the objects that can form an equilateral triangle, as in the diagram below. The n^{th} triangular number is the sum of the first n integers, as shown in the following figure illustrating the first four triangular numbers (what is the fifth one?):



Design a circuit that takes four inputs x_0, x_1, x_2, x_3 and outputs True (T or 1) if the integer value $(x_3x_2x_1x_0)_{2,4}$ is a triangular number, and False (F or 0) otherwise. You may assume that 0 is not a triangular number. (*Credit: UBC Department of Computer Science*)

5. Circuits, Compound Propositions, Logical Equivalence

1. Draw a logic circuit that uses **exactly three** gates and is logically equivalent to

$$q \leftrightarrow (p \wedge r)$$

   You may (only) use AND, OR, NOT, and XOR gates.
2. Write a compound proposition which is logically equivalent to

$$(p \oplus q) \leftrightarrow r$$

   You may only use the logical operators negation ($\neg$), conjunction ($\wedge$), and disjunction ($\vee$).

3. Find a compound proposition that is in DNF (disjunctive normal form) and is logically equivalent to

$$(p \lor q \lor \neg r) \land (p \lor \neg q \lor r) \land (\neg p \lor q \lor r)$$

6. Translating Propositional Logic

p is "The display is 13.3-inch"

q is "The processor is 2.2 GHz"

r is "There is at least 128GB of flash storage"

s is "There is at least 256GB of flash storage"

u is "There is at least 512GB of flash storage"

1. Are the statements

$$p \to (r \lor s \lor u) \quad , \quad q \to (s \lor u) \quad , \quad p \leftrightarrow q \quad , \quad \neg u$$

   consistent? If so, translate to English a possible assignment of truth values to the input propositions that makes all four statements true simultaneously.

2. Consider this statement in English:

   > It's not the case that both the display is 13.3-inch and the processor is 2.2 GHz.

   Determine whether each of the compound propositions below is equivalent to the **negation** of that statement, and justify your answers using either truth tables or other equivalences.

   Possible compound propositions:

   - $\neg p \lor \neg q$
   - $\neg(p \to \neg q)$
   - $\neg(p \land q)$
   - $(\neg p \leftrightarrow \neg q) \land p$

3. Consider the compound proposition

$$(p \land q) \to (r \lor s \lor u)$$

   Express the **contrapositive** of this conditional as a compound proposition.

   Then, give an assignment of truth values to each of the input propositional variables for which the original compound proposition is True but its **converse** is False.

7. Evaluating predicates, Evaluating nested predicates

1. Over the domain $\{1, 2, 3, 4, 5\}$ give an example of predicates $P(x), Q(x)$ which demonstrate that

$$(\forall x P(x)) \lor (\forall x Q(x)) \qquad \text{is not logically equivalent to} \qquad \forall x(\ P(x) \lor Q(x)\ )$$

2. Over the domain $\{0, 1, 2\}$, give an example of predicates $P(x), Q(x)$ for which all of these statements are true:

$$\forall x(P(x) \to Q(x))$$
$$\exists x\, P(x)$$
$$\exists x\, \neg P(x)$$
$$\exists x\, \neg Q(x)$$

8. **Evaluating predicates, Evaluating nested predicates** Recall that S is defined as the set of all RNA strands, where each strand is a nonempty string made of the bases in $B = \{\mathtt{A}, \mathtt{U}, \mathtt{G}, \mathtt{C}\}$. Recall the definition of the following predicates: $L_{\mathtt{A}}$ with domain S is defined recursively by:

   Basis step: $L_{\mathtt{A}}(\mathtt{A}) = T$, $L_{\mathtt{A}}(\mathtt{C}) = L_{\mathtt{A}}(\mathtt{G}) = L_{\mathtt{A}}(\mathtt{U}) = F$
   Recursive step: If $s \in S$ and $b \in B$, then $L_{\mathtt{A}}(sb) = L_{\mathtt{A}}(s)$

   $P_{\mathtt{AUC}}$ with domain S is defined as the predicate whose truth set is the collection of RNA strands where the string $\mathtt{AUC}$ is a substring (appears inside s, in order and consecutively)
   L with domain $S \times \mathbb{Z}^+$ is defined by, for $s \in S$ and $n \in \mathbb{Z}^+$,

$$L(s, n) = \begin{cases} T & \text{if } rnalen(s) = n \\ F & \text{otherwise} \end{cases}$$

   1. Give a witness for $\exists s_1\, \exists s_2\, (L(s_1, 4) \land L(s_2, 4) \land \neg(s_1 = s_2))$ where S is the set of RNA strands, $L(s, n)$ is a predicate with domain $S \times \mathbb{Z}^+$ that is true when s has length n
   2. Give a counterexample that disproves $\forall i\, (\neg L(\mathtt{ACU}, i))$
   3. Determine which of the following statements is True or False (you do not need to prove your answer).
      (a) $\exists n \in \mathbb{Z}^+\ \forall s \in S\ (P_{\mathtt{AUC}}(s) \to \neg L(s, n))$
      (b) $\exists n \in \mathbb{Z}^+\ \forall s \in S\ (P_{\mathtt{AUC}}(s) \land \neg L(s, n))$
      (c) $\forall s \in S\ \exists n \in \mathbb{Z}^+\ (P_{\mathtt{AUC}}(s) \lor \neg L(s, n))$
      (d) $\forall s \in S\ \exists n \in \mathbb{Z}^+\ (L(s, n) \to \neg P_{\mathtt{AUC}}(s))$

9. **Predicates and Quantifiers** Express each of the following statements symbolically using quantifiers, variables, propositional connectives, and predicates (make sure to define the domain and the meaning of any predicates you introduce). Then, express its negation as a logically equivalent compound proposition without a $\neg$ in front. Decide whether the statement or its negation is true, and prove it.

(a) If m is an integer, $m^2 + m + 1$ is odd.
(b) For all integers n greater than 5, $2^n - 1$ is not prime.
(c) If n and m are even integers, then $n - m$ is even.

10. Proof strategies Prove each of the following claims. Use only basic definitions and general proof strategies in your proofs; do not use any results proved in class / the textbook as lemmas. You may use basic properties of real numbers that we stated (without proof) in class, e.g. that the difference of two integers is an integer.

(a) The difference of any rational number and any irrational number is irrational.
(b) If a, b, c are integers and $a^2 + b^2 = c^2$, then at least one of a and b is even.
(c) For any integer n, $n^2 + 5$ is not divisible by 4.
(d) For all positive integers a, b, c, if $a \not| bc$ then $a \not| b$. (The notation $a \not| b$ means "a does not divide b").

Bonus: Watch the video `https://www.youtube.com/watch?v=MhJN9sByRS0` and write down the statement and one or both of the proofs described using the notation, definitions, and proof strategies from CSE 20.

11. Sets Prove or disprove each of the following statements.

(a) For all sets A and B, $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$.
(b) For all sets A and B, $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.
(e) For any sets A, B, C, D, if the Cartesian products $A \times B$ and $C \times D$ are disjoint then either A and C are disjoint or B and D are disjoint (or both).
(f) There are sets A, B such that $A \in B$ and $A \subseteq B$.
(g) For all sets A, B, C: $A \cap B = \emptyset$ and $B \cap C = \emptyset$ if and only if $(A \cap B) \cap C = \emptyset$.

12. Induction and Recursion Prove the following statements:

(a) $\forall s \in S\ (\ rnalen(s) = basecount(s, \texttt{A}) + basecount(s, \texttt{C}) + basecount(s, \texttt{U}) + basecount(s, \texttt{G})\ )$ where S RNA strands and the functions *rnalen* and *basecount* are define recursively as:

		$rnalen : S$	$\to \mathbb{Z}^+$
Basis Step:	If $b \in B$ then	$rnalen(b)$	$= 1$
Recursive Step:	If $s \in S$ and $b \in B$, then	$rnalen(sb)$	$= 1 + rnalen(s)$

		$basecount : S \times B$	$\to \mathbb{N}$
Basis Step:	If $b_1 \in B, b_2 \in B$	$basecount(b_1, b_2)$	$= \begin{cases} 1 & \text{when } b_1 = b_2 \\ 0 & \text{when } b_1 \neq b_2 \end{cases}$
Recursive Step:	If $s \in S, b_1 \in B, b_2 \in B$	$basecount(sb_1, b_2)$	$= \begin{cases} 1 + basecount(s, b_2) & \text{when } b_1 = b_2 \\ basecount(s, b_2) & \text{when } b_1 \neq b_2 \end{cases}$

(b) $\forall l_1 \in L\, \forall l_2 \in L\, (sum(concat(l_1, l_2)) = sum(l_1) + sum(l_2))$, where the function $concat$ is defined as:

		$concat : L \times L$	$\to L$
Basis Step:	If $l \in L$	$concat([], l)$	$= l$
Recursive Step:	If $l, l' \in L$ and $n \in \mathbb{N}$, then	$concat((n, l), l')$	$= (n, concat(l, l'))$

and the function $sum : L \to \mathbb{N}$ that sums all the elements of a list and is defined by:

		$sum : L$	$\to \mathbb{N}$
Basis Step:		$sum([])$	$= 0$
Recursive Step:	If $l \in L, n \in \mathbb{N}$	$sum((n, l))$	$= n + sum(l)$

13. Induction and Recursion At time 0, a particle resides at the point 0 on the real line. Within 1 second, it divides into 2 particles that fly in opposite directions and stop at distance 1 from the original particle. Within the next second, each of these particles again divides into 2 particles flying in opposite directions and stopping at distance 1 from the point of division, and so on. Whenever particles meet they annihilate (leaving nothing behind). How many particles will there be at time $2^{11} - 1$? You do not need to justify your answer. Hint: Derive a formula for the number/ locations of particles at time $2^n - 1$ for arbitrary positive integer n, prove the formula using induction, and apply it when $n = 11$.

14. Induction and Recursion Prove that every positive integer has a base 3 expansion. *Hint: use strong induction.*

15. Functions & Cardinalities of sets Prove each of the following claims.

(a) For the "function" $f : \mathbb{Z} \to \{0, 1, 2, 3\}$ given by $f(x) = x$ **mod** 5, it is not the case that every element of the domain maps to exactly one element of the codomain (that is, it is not a well-defined function).

(b) The "function" $f : \mathcal{P}(\mathbb{Z}^+) \to \mathbb{Z}$ given by

$$f(A) = \text{the maximum element in A}$$

is not well-defined.

(c) There is a one-to-one function with domain $\{a, b, c\}$ and codomain $\mathbb{R}$.

(d) There is an onto function with domain R and codomain $\{\pi, \frac{1}{17}\}$, where R is the set of user ratings in a database with 5 movies.

(e) The Cartesian product $\mathbb{Z}^+ \times \{a, b, c\}$ is countable.

(f) The interval of real numbers $\{x \in \mathbb{R} \mid 5 \leq x \leq 8\}$ is uncountable. *Hint: you may use the fact that the unit interval $\{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ is uncountable.*

hw1-definitions-and-notation

CSE20W26

Due: 01/15/26 at 5pm (no penalty late submission until 8am next morning)

In this assignment,

You will practice reading and applying definitions to get comfortable working with mathematical language.

Relevant class material: Week 1.

You will submit this assignment via Gradescope (https://www.gradescope.com) in the assignment called "hw1-definitions-and-notation".

For all HW assignments: These homework assignments may be done individually or in groups of up to four students. Please ensure your name(s) and PID(s) are clearly visible on the first page of your homework submission, start each question on a new page, and upload the PDF to Gradescope. If you're working in a group, *submit only one submission per group*: one partner uploads the submission through their Gradescope account and then adds the other group member(s) to the Gradescope submission by selecting their name(s) in the "Add Group Members" dialog box. You will need to re-add your group member(s) every time you resubmit a new version of your assignment.

All submitted homework for this class must be typed. You can use a word processing editor if you like (Microsoft Word, Open Office, Notepad, Vim, Google Docs, etc.) but you might find it useful to take this opportunity to learn LaTeX. LaTeX is a markup language used widely in computer science and mathematics. The homework assignments are typed using LaTeX and you can use the source files as templates for typesetting your solutions.

Integrity reminders

- Problems should be solved together, not divided up between the partners. The homework is designed to give you practice with the main concepts and techniques of the course, while getting to know and learn from your classmates.
- Use only resources explicitly allowed for each assignment. Resources not affiliated with this quarter's version of the class may use inconsistent notation or definitions. When

consulting resources, balance getting the help you need while also protecting the learning experience you will have when the 'aha' moments of solving the problem authentically happen.

- Do not share written solutions or partial solutions for homework with other students in the class who are not in your group. Doing so would dilute their learning experience and detract from their success in the class.

When evaluating your homework solutions, we will be considering:

- The correctness of any final answers.
- Whether ideas are presented clearly and logically. You should explain how you arrived at your conclusions, using mathematically sound reasoning. Whether you use formal proof techniques or write a more informal argument for why something is true, your answers should always be well-supported. Your goal should be to convince the reader that your results and methods are sound. A rule of thumb for the level of detail to include is to imagine someone who is taking the class alongside you but missed about a week's worth of content: does your writeup have enough detail for them to learn from it?

To demonstrate your honest effort in answering homework questions, we expect you to include your attempt to answer *each* part of the question. If you get stuck with your attempt, you can still demonstrate your effort by explaining where you got stuck and what you did to try to get unstuck (e.g. reviewing annotated notes from class, checking the relevant textbook sections, looking up comparable questions on the relevant weekly review quizzes, asking questions on Piazza, attending (instructor/TA/tutor) office hours, consulting generative AI tools, etc.).

Assigned questions

1. Modeling

(a) In class, we used 4-tuples to represent the user ratings for four movies. This representation is memory-efficient because we use the order of the components in the 4-tuples to represent which move is being rated. However, it is not easily extended when we want to add new movies to the database. Define a new model that would allow us to represent the user ratings of movie databases where we allow for new movies to be added. Use only the data types we have talked about in class: sets, n-tuples, and strings. Explain the design choices that you used to define your model by referencing properties of the data-type(s) you choose. Demonstrate your model by showing how the rating of the user who dislikes Superman, Wicked and likes KPop Demon Hunters, A Minecraft Movie is represented.

(b) Colors can be described as amounts of red, green, and blue mixed together [1] Mathematically, a color can be represented as a 3-tuple (r, g, b) where r represents the red component, g the green component, b the blue component and where each of r, g, b must be a value from this collection of numbers:

{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255}

(This is the same definition as in the Week 1 Review quiz.)

Can you find two different 3-tuples that represent colors that are indistinguishable to your eye? You can use the website in the footnote to play around with different choices of red, green, and blue levels to see if you can distinguish between the resulting colors. Why or why not?

A complete answer will include the specific example 3-tuples that work, along with a description of the colors that they represent and why they are indistinguishable, or an explanation of why there can't be such an example.

2. Sets and functions

   (a) Each of the sets below is described using set builder notation or recursion or as a result of set operations applied to other known sets. Rewrite each of the sets using the roster method.

   Remember our discussions of data-types: use clear notation that is consistent with our class notes and definitions to communicate the data-types of the elements in each set.

Sample response that can be used as reference for the detail expected in your answer:

The set $\{\texttt{A}\} \circ \{\texttt{AU}, \texttt{AC}, \texttt{AG}\}$ can be written using the roster method as

$$\{\texttt{AAU}, \texttt{AAC}, \texttt{AAG}\}$$

because set-wise concatenation gives a set whose elements are all possible results of concatenating an element of the left set with an element of the right set. Since the left

[1] This RGB representation is common in web applications. Many online tools are available to play around with mixing these colors, e.g. `https://www.w3schools.com/colors/colors_rgb.asp`.

set in this example only has one element, namely `A`, each of the elements of the set we described starts with `A`. There are three elements of this set, one for each of the distinct elements of the right set.

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i.
$$\{n \in \mathbb{Z}^+ \mid n \leq 3\} \times \{m \in \mathbb{N} \mid m \leq 3\}$$

ii. The set X defined recursively as

Basis Step:	$1 \in X, 3 \in X, 5 \in X$
Recursive Step:	If the integer $n \in X$, then the result of multiplying n by -1 is in X

iii.
$$\{x \in S \mid rnalen(x) = 2\} \circ \{x \in S \mid rnalen(x) = 0\}$$

where S is the set of RNA strands and *rnalen* is the recursively defined function that we discussed in class,

		$rnalen : S$	$\to \mathbb{Z}^+$
Basis Step:	If $b \in B$ then	$rnalen(b)$	$= 1$
Recursive Step:	If $s \in S$ and $b \in B$, then	$rnalen(sb)$	$= 1 + rnalen(s)$

iv.
$$\{(r, g, b) \in C \mid r + g + b = 2 \text{ and } g = 1\}$$

where $C = \{(r, g, b) \mid 0 \leq r \leq 255, 0 \leq g \leq 255, 0 \leq b \leq 255, r \in \mathbb{N}, g \in \mathbb{N}, b \in \mathbb{N}\}$.

(b) Recall the function which takes an ordered pair of ratings 4-tuples and returns a measure of the difference between them

$$d_0 : \{-1, 0, 1\}^4 \times \{-1, 0, 1\}^4 \to \mathbb{R}$$

given by

$$d_0(\ (\ (x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)\ )\ ) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + (x_4 - y_4)^2}$$

Define a **new** function which we'll call d_{new} with the same domain $\{-1, 0, 1\}^4 \times \{-1, 0, 1\}^4$ and codomain $\mathbb{R}$ but where there is some example pair of ratings 4-tuples

$$(\ (x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)\ )$$

where

$$d_0(\ (\ (x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)\ )\ ) \neq d_{new}(\ (\ (x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)\ )\ )$$

Your answer should include **both** a precise and clear definition for the rule defining d_{new} which unambiguously specifies output for each input of the function **and** the example ordered pair of ratings 4-tuples that demonstrate that the functions are not equal. Also include a justification of your answers with (clear, correct, complete) calculations for each of the function applications and/or references to definitions and connecting them with the desired conclusion.

(c) A function *basecount* that computes the number of a given base b appearing in a RNA strand s is defined recursively:

$$
\begin{array}{lll}
 & basecount: S \times B & \to \mathbb{N} \\
\text{Basis Step:} & & \\
\text{If } b_1 \in B, b_2 \in B & basecount(\ (b_1, b_2)\ ) & = \begin{cases} 1 & \text{when } b_1 = b_2 \\ 0 & \text{when } b_1 \neq b_2 \end{cases} \\
\text{Recursive Step:} & & \\
\text{If } s \in S, b_1 \in B, b_2 \in B & basecount(\ (sb_1, b_2)\ ) & = \begin{cases} 1 + basecount(\ (s, b_2)\ ) & \text{when } b_1 = b_2 \\ basecount(\ (s, b_2)\ ) & \text{when } b_1 \neq b_2 \end{cases}
\end{array}
$$

Consider the function application

$$basecount(\ (\texttt{ACAU}, \texttt{A})\ )$$

What is the input? What is the output? Give an example of a different choice of input that gives the same output.

Your answer should include clearly labeled answers to each of the three parts of the question, along with a justification for the values of the applications that makes specific reference to the parts of the recursive definition of the *basecount* function used to calculate it.